

Grade 1

Ontario Curriculum

Spatial Sense

Map It, Move It!



Compass Coco the Cat

Unit Overview: Map It, Move It!

Grade: Grade 1 | **Strand:** Spatial Sense

Target Expectations: E1.1, E1.2, E1.3, E1.4, E1.5

Learning Goal: Children will sort shapes by attribute, build 3D structures, identify 2D shapes within them, make shapes with matching halves, describe locations, and give/follow directions.

Lesson	Lesson Title	Learning Goal
Day 1	Sort My Shapes	I can sort shapes by one rule and name the rule.
Day 2	Build and Find	I can build a 3D structure and find 2D shapes inside it.
Day 3	Matching Halves	I can make shapes with matching halves.
Day 4	Where Is It?	I can describe where something is using position words.
Day 5	Treasure Trail Capstone	I can give and follow directions to move.

Lesson 1: Sort My Shapes

Goal: We sort shapes by ONE attribute. Same shapes can be sorted in DIFFERENT ways.

Expectation: E1.1

Learning Goal: "I can sort shapes by one rule and name the rule."

Materials: M1_shape_set, M7_attribute_chart, M10_unit_poster

1. Minds On (12 mins): Tig Sorts the Trail Pile

Tig: I found a PILE of shapes on the trail. Help me SORT them. What ONE attribute can I use? Tig opens: 'Pick ONE! SAY it! SORT it!' (Class echoes.)

Bring children to the carpet. Tig and Coco puppets. 'Friends — I'm Trekker Tig the Tiger Cub, and this is my friend Compass Coco the Cat. Today we sort shapes by ATTRIBUTES.'

2. Action (25 mins): Sort Stations

- Whole group: sort 4 piles.
- Three stations.
- Stretch challenge at Station C: sort by 2 attributes (red AND big).
- Discuss attribute choice.
- Circulate to record diagnostic evidence on the sort tracker.
- INVENT-AND-COMPARE: Pairs pick their OWN attribute (no teacher list) and sort 8 shapes.

3. Consolidation (13 mins): Tig's Sort Brag

Discuss:

- What is an ATTRIBUTE?
- Can the same pile be sorted in DIFFERENT ways?
- How do you NAME your rule?
- Why do detectives sort?
- Tig's preview: tomorrow we BUILD with shapes (Lesson 2 — E1.2).

Exit: Tig's three-step refrain — class chants 3x: 'Pick ONE!'

Lesson 2: Build and Find

Goal: We can BUILD 3D structures. Inside every structure are 2D SHAPES.

Expectation: E1.2

Learning Goal: "I can build a 3D structure and find 2D shapes inside it."

Materials: M2_3d_blocks, M7_attribute_chart, M10_unit_poster

1. Minds On (12 mins): Tig Builds a Cabin

Tig: I built a CABIN out of 3D blocks. Look — I see SQUARES on the walls and a TRIANGLE on the roof! Tig opens: 'Turn it! Find it! Name it!'

Bring children to the carpet. Tig and Coco puppets. 'Friends — yesterday we sorted shapes. Today we BUILD with 3D blocks AND find 2D shapes inside.'

2. Action (25 mins): Build Stations

- Whole group: build a 3-block tower.
- Three stations.
- Stretch challenge at Station C: build with 5 different 3D shapes.
- Discuss the face-counting rule.
- Circulate to record formative evidence on the build tracker.
- STRATEGY COMPARE: Two pairs share HOW they made sure they counted every face — Pair A 'I turned 4 times then flipped'; Pair B 'I marked each face with a finger.' Class debates...

3. Consolidation (13 mins): Tig's Build Brag

Discuss:

- What is a STRUCTURE?
- What 2D shapes are on a CUBE?
- What 2D shapes are on a CYLINDER?
- Did you find any surprising shapes?

Exit: Tig + Coco refrain — class chants: 'Turn it!'

Lesson 3: Matching Halves

Goal: A shape with MATCHING HALVES has a line of SYMMETRY. Each half is a MIRROR of the other.

Expectation: E1.3

Learning Goal: "I can make shapes with matching halves."

Materials: M3_symmetry_set, M10_unit_poster

1. Minds On (12 mins): Coco Folds the Maple Leaf

Coco: I FOLDED a maple leaf. Look — both halves MATCH! It has a line of SYMMETRY.

Bring children to the carpet. Tig and Coco puppets. 'Friends — yesterday we built. Today we look at MATCHING HALVES.'

2. Action (25 mins): Symmetry Stations

- Whole group: fold 4 shapes.
- Three stations.
- Stretch challenge at Station C: make a shape with TWO lines of symmetry (like a square or rectangle).
- Discuss the fold test.
- Circulate to record formative evidence on the symmetry tracker.

3. Consolidation (13 mins): Coco's Symmetry Brag

Discuss:

- What are MATCHING HALVES?
- How do we test for matching halves?
- What's a Canadian shape with matching halves?
- What's a shape with NO matching halves?

Exit: Each child names ONE shape with matching halves and ONE without.

Lesson 4: Where Is It?

Goal: We describe LOCATION with position words: above, below, beside, between, in front, behind.

Expectation: E1.4

Learning Goal: "I can describe where something is using position words."

Materials: M4_treasure_map, M5_treasure_tokens, M8_position_chart, M10_unit_poster

1. Minds On (12 mins): Tig Hides the Treasures

Tig: I HID treasures on the map! Use position words to tell me WHERE they are. Above the cabin? Beside the lake?

Bring children to the carpet. Tig and Coco puppets. 'Friends — yesterday matching halves. Today we describe LOCATION with position words.'

2. Action (25 mins): Position Stations

- Whole group: describe 6 hidden treasures on the class map.
- Three stations.
- Stretch challenge at Station C: write a clue using TWO position words ('above and beside').
- Discuss the frame.
- Circulate to record formative evidence on the position tracker.

3. Consolidation (13 mins): Coco's Position Brag

Discuss:

- What does LOCATION mean?
- What's the difference between BESIDE and BETWEEN?
- What position word do you use for 'on top'?
- Where is your bag right now?

Exit: Each child describes ONE classroom item: 'The ___ is ___ the ___.'

Lesson 5: Treasure Trail Capstone

Goal: We can GIVE and FOLLOW directions to move from one place to another.

Expectation: E1.5

Learning Goal: "I can give and follow directions to move."

Materials: M1_shape_set, M2_3d_blocks, M3_symmetry_set, M4_treasure_map

1. Minds On (12 mins): Coco Gives Directions

Coco: I'll tell you how to find the treasure! Forward 3, turn right, forward 2. FOLLOW me! Tig opens: 'Map it, MOVE it!' Coco: 'Verb plus NUMBER!' Class: 'every TIME!'

Bring children to the carpet. Tig and Coco puppets. 'Friends — today is our CAPSTONE. We use ALL we know: SORT, BUILD, MATCH, LOCATE, MOVE!'

2. Action (25 mins): The Treasure Trail

- RECAP (5 min): Tig + Coco model each skill on the carpet — SORT, BUILD, MATCH, LOCATE — one quick example per skill.
- STATIONS (20 min): children rotate through 4 capstone stations and record on the template.
 - DIRECT: walk treasure trail with a partner.
 - BUILD: sketch 2D shapes inside a...

3. Consolidation (13 mins): Tig's Capstone Brag

Discuss:

- Which station was your favourite?
- What's a DIRECTION?
- How do you describe where something is?
- What did you build?

Exit: Tig + Coco — Tig: 'Map it, MOVE it!' Coco: 'Verb + NUMBER!' Class: 'every TIME!' Repeat 3x.

Name: _____

Date: _____

Tig's Sort Sheet

Learning Goal: I can sort shapes by one rule and name the rule.

Instructions for the Student:

1. Look at the pile.
2. Sort the same 8 shapes by SHAPE.
3. Pick an attribute (size, sides, etc.).

Part 2: Same shapes, NEW rule. Sort by SHAPE.

Sort the same 8 shapes by SHAPE. Draw 3 piles.

Name: _____

Date: _____

Tig's Build Sheet

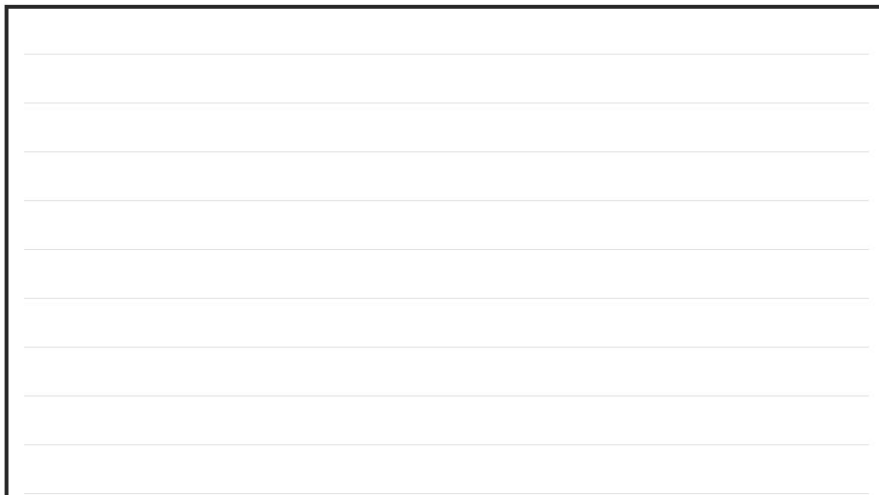
Learning Goal: I can build a 3D structure and find 2D shapes inside it.

Mission

1. Look at each 3D structure.
2. Imagine a tower of 4 cubes.
3. Build (or draw) any structure.

Part 3: Build any structure.

Build (or draw) any structure. Find 3+ different 2D shapes inside.

A large rectangular box with a black border, containing ten horizontal grey lines for drawing a 3D structure.

Name: _____

Date: _____

Coco's Symmetry Page

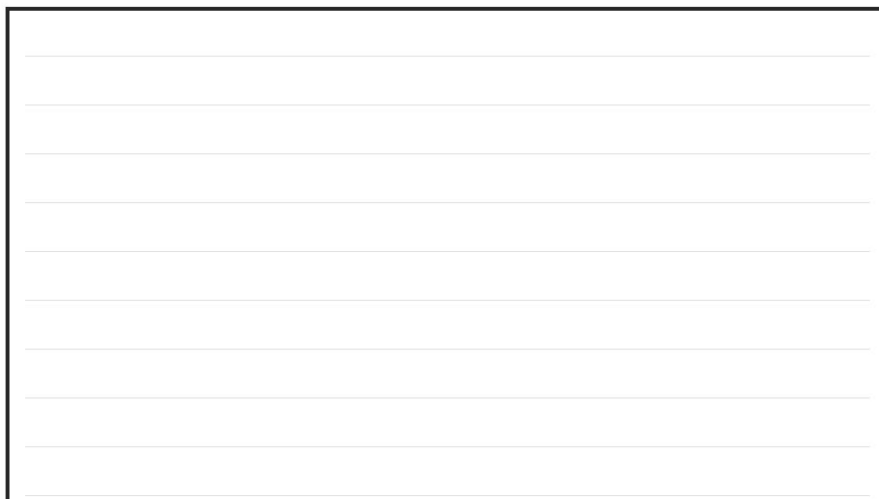
Learning Goal: I can make shapes with matching halves.

Instructions for the Student:

1. Circle the shapes that have MATCHING HALVES.
2. Draw a line of symmetry on each shape.
3. Draw a shape that has MATCHING HALVES.

Part 3: Make your own.

Draw a shape that has MATCHING HALVES. Show the line of symmetry.

A large rectangular box with a black border, containing ten horizontal grey lines for drawing.

Date: _____

Learning Goal: I can describe where something is using position words.

1. Look at each map.
2. Read each clue.
3. Pick 3 things in your bedroom.

[illegible]

Name: _____

Date: _____

Tig's Treasure Trail

Learning Goal: I can give and follow directions to move.

Instructions for the Student:

1. Pick a rule.
2. Draw a line of symmetry on each shape.
3. Look at the trail map.

Part 3: MOVE!

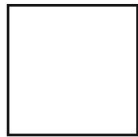
Look at the trail map. Write 3 directions to find the treasure.

Shape Set 2D + 3D

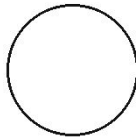
(M1_shape_set)

A set of 30 paper shape tokens (mix of 2D and 3D) for sorting by attribute.

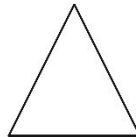
2D Shapes



square



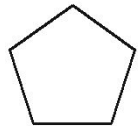
circle



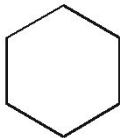
triangle



rectangle



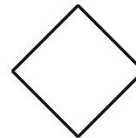
pentagon



hexagon



trapezoid



rhombus

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 5

Prep: ~25 min

TEACHER PREP

1. Print 10 sets total (1 per pair).
2. Laminate before cutting.
3. Cut along the grid lines into 30 individual tokens per set.
4. Store each set in a labelled zip-top bag.

Paper 3D Blocks

(M2_3d_blocks)

A set of 12 paper-fold 3D blocks (cubes, cones, cylinders, pyramids, prisms) for building structures.

Paper 3D Blocks	
1	Cone Unfolds
2	Cylinder Rolls
3	Pyramid Bases
4	Tabs For Assembly

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 3

Laminate: no

USE IN CLASS

Quantity: 10 sets

Lessons: 2, 5

Prep: ~60 min

TEACHER PREP

1. Print 6 sets (1 per pair plus 2 extras).

2. Cut along solid lines and fold along dashed lines.

3. Tape or glue tabs to assemble each block.

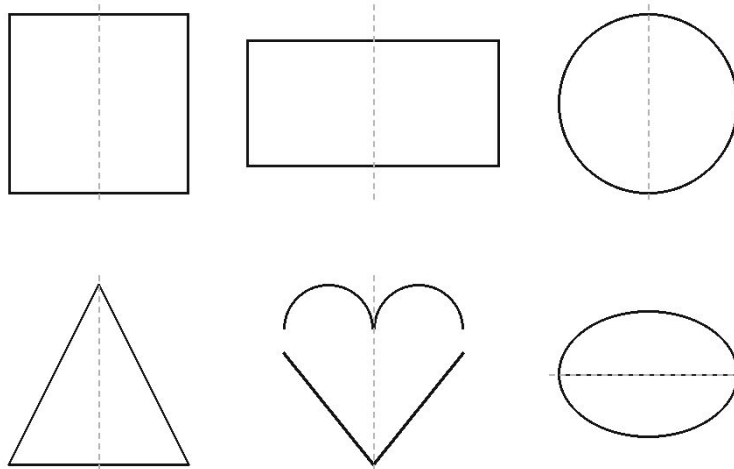
4. Store each pair's set in a labelled box.

Symmetry Shape Set

(M3_symmetry_set)

A set of 16 paper shape halves for matching-halves activities.

Symmetry — Mirror Lines



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 10 sets

Lessons: 3, 5

Prep: ~20 min

TEACHER PREP

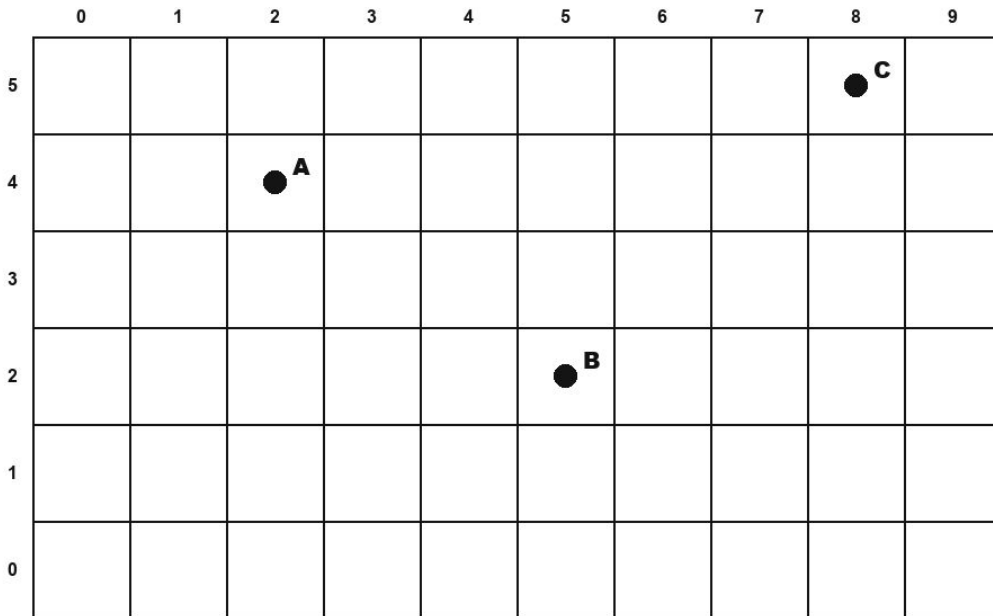
1. Print 10 sets (1 per pair).
2. Laminate before cutting.
3. Cut into 16 individual half-shapes per set.
4. Store each set in a labelled bag.

Treasure Map Mat

(M4_treasure_map)

A letter-size treasure map with grid cells for placing tokens and giving directions.

Coordinate Grid (10 × 6)



PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 12 mats (1 per pair plus 2 extras).

2. Laminate for repeated dry-erase use.

3. Distribute one mat per pair.

USE IN CLASS

Quantity: 10-12 mats

Lessons: 4, 5

Prep: ~15 min

Treasure Tokens

(M5_treasure_tokens)

A set of 12 small treasure tokens (gold, silver, gem, key) for hiding on the map.

Treasure Tokens



PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 10 sets (1 per pair).
2. Laminate before cutting.
3. Cut into 12 tokens per set.
4. Store with the Treasure Map Mat.

USE IN CLASS

Quantity: 10 sets
Lessons: 4, 5
Prep: ~15 min

Treasure Trail Template

(M6_capstone_template)

A letter-size capstone template with map, direction lines, and reflection.

Treasure Trail Template	
1	START square _____
2	TREASURE square on a 6×4 grid _____
3	Build _____
4	Match _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

TEACHER PREP

1. Print 25 templates (1 per child plus 5 extras).

2. Stack near the teacher table.

3. No lamination needed.

USE IN CLASS

Quantity: 25 templates

Lessons: 5

Prep: ~5 min

Attribute Anchor Chart

(M7_attribute_chart)

A teacher-display anchor chart showing 6 attributes for sorting.

Shape Attributes
• Sides — straight or curved? • Corners — how many? • Equal sides? • Right angles? • Symmetry — does it match?

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to the front classroom wall before Lesson 1.

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 5

Prep: ~10 min

Position Words Chart

(M8_position_chart)

A teacher-display anchor chart of 6 position words.

Position Words

LEFT	RIGHT	UP	DOWN
ABOVE	BELOW	INSIDE	OUTSIDE

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape next to Attribute Chart on the classroom wall.

USE IN CLASS

Quantity: 1 chart

Lessons: 4, 5

Prep: ~10 min

Direction Cards

(M9_direction_cards)

A set of 12 direction cards (forward, backward, left, right, up, down
— with numbers).

Direction Cards	
1	Forward
2	Backward
3	Up
4	Down

PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 10 sets (1 per pair).
2. Laminate before cutting.
3. Cut into 12 cards per set.
4. Store each set in a labelled bag.

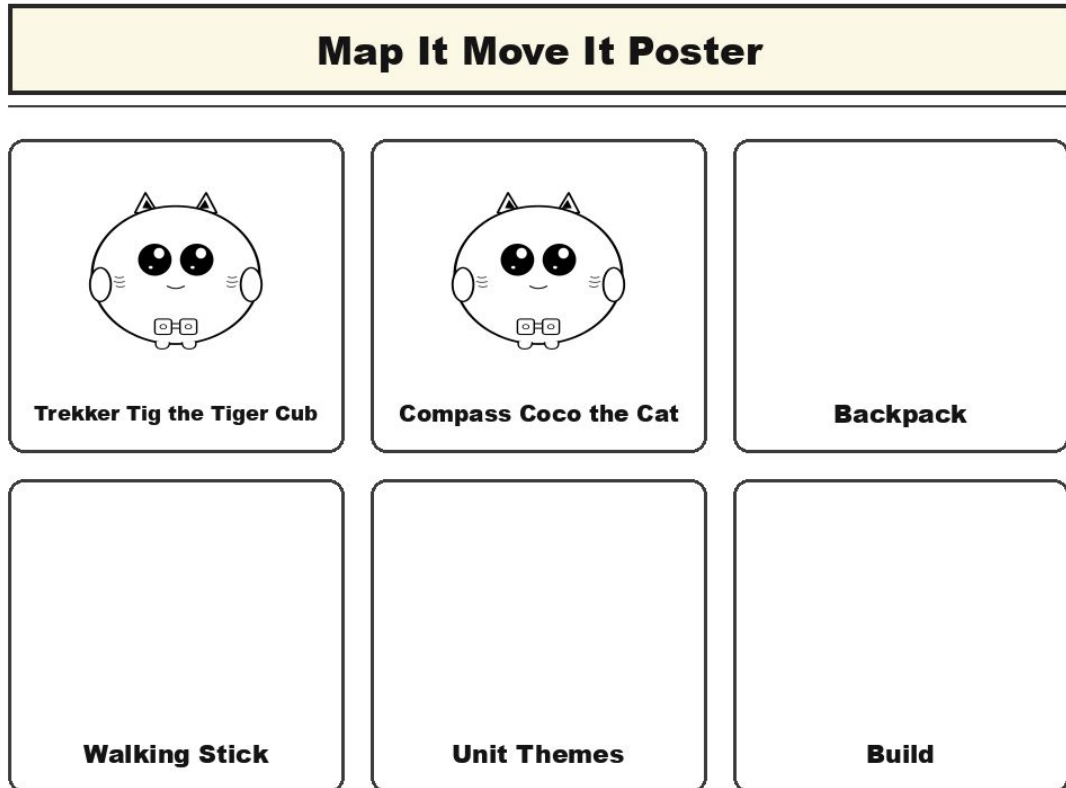
USE IN CLASS

Quantity: 10 sets
Lessons: 5
Prep: ~15 min

Map It Move It Poster

(M10_unit_poster)

A unit anchor poster of Tig and Coco with sample shapes and directions.



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 1 copy on letter or 11x17.
2. Laminate.
3. Tape to the front classroom wall on Day 1.

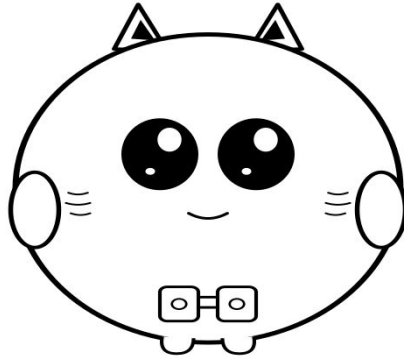
USE IN CLASS

Quantity: 1 poster
Lessons: 1, 2, 3, 4, 5
Prep: ~10 min

Trekker Tig Puppet

(char_tig_puppet)

A paper puppet of Trekker Tig the Tiger Cub used by the teacher in the Minds-On phase of every lesson.



Trekker Tig the Tiger Cub

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

TEACHER PREP

1. Print 1 copy of Tig on cardstock if possible.

2. Cut around the outside of Tig's silhouette including the tab.

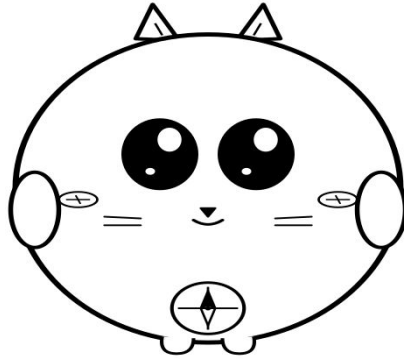
3. Laminate before assembly.

4. Glue or tape a popsicle stick behind the bottom tab.

Compass Coco Puppet

(char_coco_puppet)

A paper puppet of Compass Coco the Cat used by the teacher in Lessons 3-5 for direction-giving demos.



Compass Coco the Cat

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 3, 4, 5

Prep: ~12 min

TEACHER PREP

1. Print 1 copy of Coco on cardstock if possible.

2. Cut around the outside of Coco's silhouette including the tab.

3. Laminate before assembly.

4. Glue or tape a popsicle stick behind the bottom tab.

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
E1.1 sort three-dimensional objects and two-dimensional shapes according to one attribute at a time, and identify the sorting rule	Sorts inconsistently. Drifts attribute mid-sort. Cannot articulate a rule.	Sorts by one attribute with prompts. Names the rule with a frame.	Independently picks an attribute, sorts consistently, and names the rule clearly.	Re-sorts the same set by a different attribute and explains how the rule changes the groupings.
E1.2 construct three-dimensional objects, and identify two-dimensional shapes contained within structures and objects	Stacks blocks randomly. Cannot identify 2D shapes on faces.	Builds simple tower. Names 1-2 shapes on visible faces.	Builds a multi-block structure and names 3+ different 2D shapes inside (rotates structure to find faces).	Counts faces, edges, vertices on a 3D shape AND identifies all 2D shapes from every angle.
E1.3 construct and describe two-dimensional shapes and three-dimensional objects that have matching halves	Cannot identify matching halves. Folds randomly without checking match.	Identifies matching halves with prompts. Sometimes finds the right fold line.	Independently identifies symmetric shapes, finds the line of symmetry by folding, and creates one matching-halves shape.	Identifies multiple lines of symmetry on shapes that have them (square = 4 lines).
E1.4 describe the relative locations of objects or people, using positional language	Uses one position word for many ('over' for everything). Cannot use the frame.	Uses 2-3 position words. Uses the frame with prompts.	Independently uses 5+ position words and the 'The ___ is ___ the ___' frame.	Combines two position words ('above and beside') and uses position vocabulary spontaneously across the day.
E1.5 give and follow directions for moving from one location to another	Gestures vaguely. Cannot give numbered directions. Cannot follow a partner's directions.	Uses words but no numbers. Follows simple 1-step directions.	Independently gives verb + number directions ('forward 3'). Follows a 3-step sequence to the target.	Plans a multi-step route in advance and writes a complete sequence.

Junior Spatial Detective Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Map It, Move It! unit and demonstrating the skills of a Junior Spatial Detective. A spatial certificate. Trekker Tig the Tiger Cub congratulates the student.

By completing this unit, this student has demonstrated:

- Sorting shapes by attribute
- Building 3D structures and finding 2D shapes
- Making shapes with matching halves
- Describing locations and giving directions

Marketplace Listing

A 5-day Grade 1 unit on spatial sense framed around Trekker Tig the Tiger Cub and Compass Coco the Cat.

Price: \$8.99 CAD

Pages: ~45

Classroom time: 250 min

Teacher prep: ~220 min

Top tags: Grade 1, math, spatial, geometry, shapes, 2D, 3D, symmetry

What's included:

- Unit blueprint with curriculum mapping (1 file)
- 5 lesson plans with minds-on, action, consolidation, expanded walkthroughs (5 files)
- 5 student worksheets with answer keys (5 files)
- Manipulatives plan with 12 print-ready assets (1 file)
- Formative + reflection sheets (1 file)
- Assessment suite (1 file: trackers, summative rubric, task script, Junior Spatial Detective certificate)